CSE 641 CV Project Report Week-5
Project Name: Athletic Readiness: Data Generation

Group Name: Deep Vision

Objective

For this week our aim was at training the Variational Autoencoder (VAE) model on the already prepared preprocessed dataset that had already been created in the last week and assessing the quality of the produced synthetic data. Our group was centered on checking whether the produced data maintain the same correlation and distribution like the original data and the correlation of the features.

Data Collection

The dataset used for training the VAE consists of 11 feature columns and 524 records. This dataset was preprocessed and cleaned during the last week, ensuring it was suitable for training generative models.

Methodology: VAE Implementation

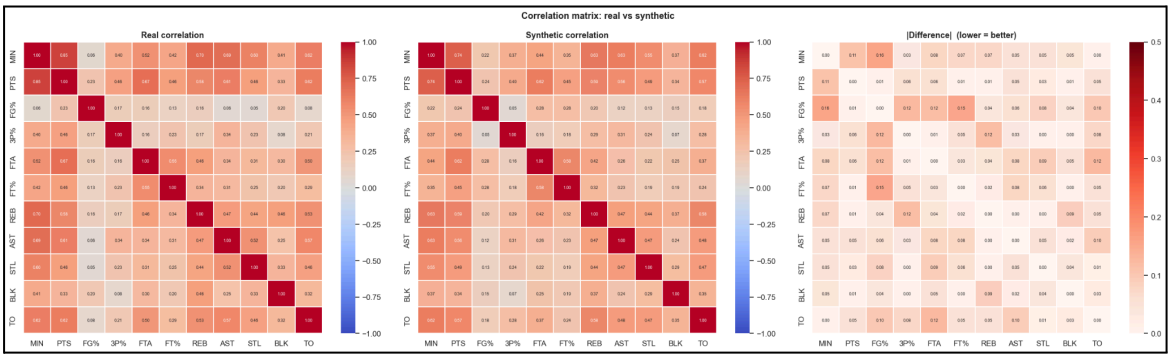
The VAE model was trained using the data to learn the underlying data distribution and sample synthesis. The encoder transformed input features into the latent space distribution and the decoder recovered the data based on the sampled latent variables.

Synthetic data samples were then created based on the learned latent space representation after the training.

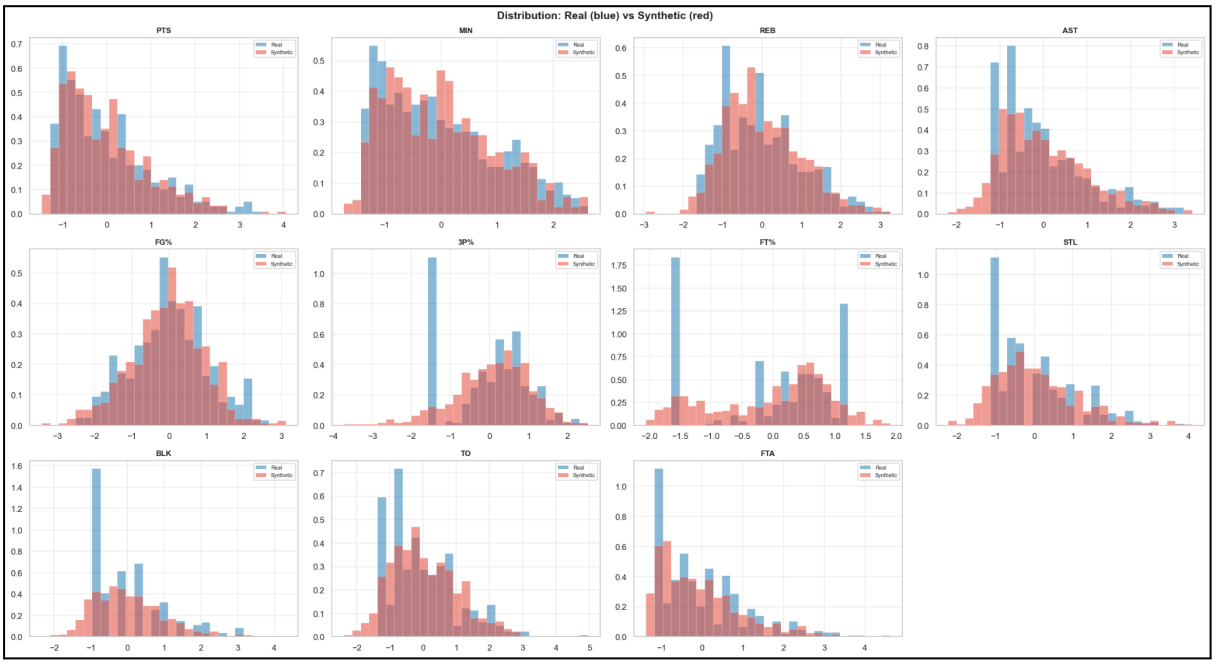
Evaluation of Generated Data

- **Correlation Analysis**

  A correlation heatmap was plotted for both the real and synthetic datasets to compare feature relationships. The heatmaps showed almost identical correlation patterns between features. The difference of covariance between the original and generated data was very less. This confirmed that the VAE preserved inter-feature dependencies, which is critical for realistic synthetic data generation.



- **Feature Distribution Analysis**
  To evaluate how well the VAE captured the original data distribution, we plotted histograms of each feature column for both the real and generated datasets on top of each other. As we can see in the images the distributions of generated features closely matched those of the original dataset. This indicates that the VAE successfully learned the marginal distributions of individual features.



The selected feature columns for generating the dataset are:

Column	Meaning
MIN	Average Minutes Played per Game
PTS	Points per Game
FG%	Field Goal Percentage
3P%	3-Point Shooting Percentage
FTA	Free Throws Attempted per Game
FT%	Free Throw Percentage
REB	Rebounds per Game
AST	Assists per Game
STL	Steals per Game
BLK	Blocks per Game
TO	Turnovers per Game

Results & Observations

- The VAE was able to produce synthetic data, which look almost similar to the original data.
- The feature-wise distribution and structure of correlations were both well-maintained.
- The model was found to have a significant ability to learn the statistical characteristics of the tabular data of athletes performance.

Conclusion

The VAE model was found to be useful in the production of quality synthetic data. The distributions and correlation are also similar with both real and generated datasets which ensures the strength of the approach. This move is a successful step towards leaving theoretical knowledge (Week 4) and embarking on practical application.

Plan for Next Week

- Implement GAN (Generative Adversarial Network) for comparison
- Evaluate VAE vs GAN performance
- Perform quantitative evaluation metrics (e.g., KL divergence, statistical similarity scores)